

DAY-3

Day 3: DNS Server Configuration Theory (30 min):How DNS resolves domain names to
Ips Practical (30 min):Configure a DNS server and test domain resolution Task:
Document the configuration and troubleshooting steps.

[Q-1] Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.

○ Objective

- Install a DNS server and create a DNS zone named **musicmania.in**.

○ Configuration Steps (BIND on Linux)

○ Step 1: Install BIND

- `sudo apt update`
- `sudo apt install bind9 bind9utils bind9-doc`

○ Step 2: Create a Zone Entry

- `sudo nano /etc/bind/named.conf.local`

□ Add:

```
zone "musicmania.in" {  
    type master;    file  
    "/etc/bind/db.musicmania.in"; };
```

○ Step 3: Create the Zone File

- `sudo cp /etc/bind/db.local /etc/bind/db.musicmania.in` Edit:
`sudo nano /etc/bind/db.musicmania.in` Example
configuration:
\$TTL 604800
@ IN SOA ns.musicmania.in. admin.musicmania.in. (
 2
 604800
 86400
 2419200
 604800)

```
@      IN NS   ns.musicmania.in.  
ns     IN A    192.168.1.50  
www    IN A    192.168.1.50
```

○ Step 4: Restart DNS Service

- `sudo systemctl restart bind9`

```
student@ubuntu:~$ sudo systemctl restart bind9  
student@ubuntu:~$ sudo systemctl status bind9  
● bind9.service - BIND Domain Name Server  
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)  
   Active: active (running) since Wed 29 May 2024 10:45:12 AM IST; 3s ago  
     Docs: man:named(8)  
    Main PID: 1152 (named)  
      Tasks: 6 (limit: 2271)  
    Memory: 12.6M  
       CPU: 58ms  
    CGroup: /system.slice/bind9.service  
            └─1152 /usr/sbin/named -f -u bind  
  
May 29 10:45:12 ubuntu systemd[1]: Starting BIND Domain Name Server...  
May 29 10:45:12 ubuntu systemd[1]: Started BIND Domain Name Server.  
student@ubuntu:~$
```

[Q-2] Add an A record in your DNS server configuration to map `www.musicmania.in` to the IP address `192.168.1.50`, then use the `nslookup` or `dig` command to verify the mapping works.

○ Add an A Record and Verify

○ A Record

- `www IN A 192.168.1.50`

○ Verify Using `nslookup`

- `nslookup www.musicmania.in`

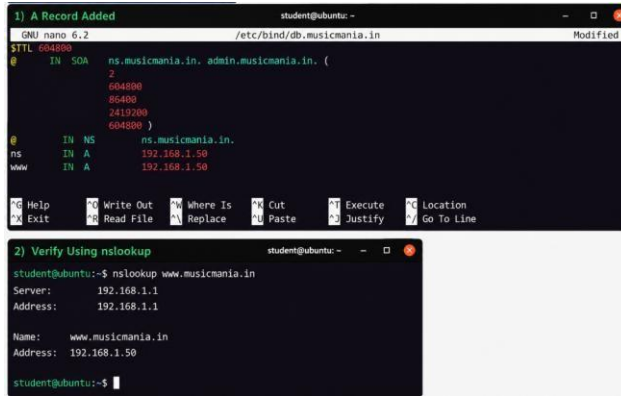
○ Sample Output

- Server: 192.168.1.1 Address:
192.168.1.1

```
Name: www.musicmania.in  
Address: 192.168.1.50
```

○ Explanation

- The A record maps the hostname www.musicmania.in to the IP address **192.168.1.50**. When users enter the domain name, the DNS server returns this IP address.



```
1) A Record Added
student@ubuntu: ~
GNU nano 6.2 /etc/bind/db.musicmania.in
#
# Generated by nano
#
# IN SOA ns.musicmania.in. admin.musicmania.in. (
# 2
# 604800
# 604800
# 2419200
# 604800 )
#
# IN NS ns.musicmania.in.
ns IN A 192.168.1.50
www IN A 192.168.1.50

2) Verify Using nslookup
student@ubuntu: ~
student@ubuntu:~$ nslookup www.musicmania.in
Server: 192.168.1.1
Address: 192.168.1.1

Name: www.musicmania.in
Address: 192.168.1.50
student@ubuntu:~$
```

[Q-3] Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.

○ Add a CNAME Record ○ Configuration

- Add the following line to the zone file:
- play IN CNAME www.musicmania.in.

○ Restart DNS

- sudo systemctl restart bind9

○ Verify

- nslookup play.musicmania.in

○ Sample Output

- Name: play.musicmania.in
Canonical name: www.musicmania.in

Name: www.musicmania.in
Address: 192.168.1.50

○ Explanation

- A CNAME (Canonical Name) record creates an alias. Here, **play.musicmania.in** points to www.musicmania.in, which then resolves to **192.168.1.50**.

```
1) Add CNAME Record (Edit Zone File) student@ubuntu: ~
GNU nano 6.2 /etc/bind/db.musicmania.in Modified
$TTL 604800
@      IN  SOA  ns.musicmania.in. admin.musicmania.in. (
        2
        604800
        86400
        2419200
        604800 )
@      IN  NS   ns.musicmania.in.
ns     IN  A    192.168.1.50
www    IN  A    192.168.1.50
play   IN  CNAME www.musicmania.in.

2) Restart DNS student@ubuntu: ~
student@ubuntu:~$ sudo systemctl restart bind9
student@ubuntu:~$ systemctl status bind9
● bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed May 29 11:05:21 IST 2024; 3s ago
     Docs: man:named(8)

3) Verify student@ubuntu: ~
student@ubuntu:~$ nslookup play.musicmania.in
Server:          192.168.1.1
Address:         192.168.1.1#53

Name:   play.musicmania.in
Canonical name: www.musicmania.in
Name:   www.musicmania.in
Address: 192.168.1.50
student@ubuntu:~$
```

[Q-4] Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.

Hint: Include at least three different troubleshooting commands or checks in your documentation.

○ Problem

The domain **musicmania.in** is not resolving correctly.

○ Step-by-Step Troubleshooting

○ Step 1: Check DNS Configuration

- Verify that the zone configuration is correct.
- Command:
 - `sudo named-checkconf`
- Expected Result:
- No errors should be displayed.

○ Step 2: Verify Zone File

- Check for syntax errors in the zone file.
- Command:

- `sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in`
- Expected Result:
- OK

○ Step 3: Restart the DNS Service

- After making changes, restart BIND.
- Command:
- `sudo systemctl restart bind9`
- Check service status:
- `sudo systemctl status bind9`
- Expected Result:
- The service should show **active (running)**.

○ Step 4: Test Domain Resolution

- Use **nslookup**.
- `nslookup www.musicmania.in`
- or use **dig**.
- `dig www.musicmania.in`
- Expected Result:
- `www.musicmania.in Address: 192.168.1.50`

○ Step 5: Check Log Files

- If the problem continues, inspect DNS logs.
- `sudo journalctl -u bind9`
- or
- `sudo tail -f /var/log/syslog`
- Look for errors such as:
- Zone file syntax error
- Permission denied
- Service failed to start

○ Step 6: Verify Firewall (Optional)

- Ensure DNS traffic is allowed. • `sudo ufw status`
- If necessary:
- `sudo ufw allow 53`

1) Check DNS Configuration

```
student@ubuntu:~$ sudo named-checkconf
student@ubuntu:~$ █
```

Expected Result: No errors should be displayed.

2) Verify Zone File

```
student@ubuntu:~$ sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
zone musicmania.in/IN: loaded serial 2
OK
student@ubuntu:~$ █
```

Expected Result: OK

3) Restart the DNS Service & Check Status

```
student@ubuntu:~$ sudo systemctl restart bind9
student@ubuntu:~$ sudo systemctl status bind9
● bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed May 29 11:15:45 IST 2024; 2s ago
     Docs: man:named(8)
    Main PID: 1234 (named)
      Tasks: 6 (limit: 2277)
     Memory: 12.1M
    CGroup: /system.slice/bind9.service
            └─1234 /usr/sbin/named -f -u bind

student@ubuntu:~$ █
```

Expected Result: The service should show active (running).

4) Test Domain Resolution (nslookup & dig)

```
student@ubuntu:~$ nslookup www.musicmania.in
Server: 192.168.1.1
Address: 192.168.1.1#53

Name: www.musicmania.in
Address: 192.168.1.50
student@ubuntu:~$ █
```

```
student@ubuntu:~$ dig www.musicmania.in
; <<>> DiG 9.18.1-ubuntu1.1-Ubuntu <<>> www.musicmania.in
;; global options: +cmd
;; Got answer:
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 12345
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 1, ADDITIONAL: 1
;; QUESTION SECTION:
;www.musicmania.in.      IN      A
;; ANSWER SECTION:
www.musicmania.in.      604800  IN      A      192.168.1.50
;; SERVER: 192.168.1.1#53(192.168.1.1)
;; WHEN: Wed May 29 11:16:30 IST 2024
;; MSG SIZE rcvd: 81

student@ubuntu:~$ █
```

Expected Result: www.musicmania.in
Address: 192.168.1.50

5) Check Log Files

```
student@ubuntu:~$ sudo journalctl -u bind9 -n 10
May 29 11:15:45 ubuntu named[1234]: zone musicmania.in/IN: loaded serial 2
May 29 11:15:45 ubuntu named[1234]: all zones loaded
May 29 11:15:45 ubuntu named[1234]: running
May 29 11:15:45 ubuntu systemd[1]: Started BIND Domain Name Server.
student@ubuntu:~$ █
```

```
student@ubuntu:~$ sudo tail -f /var/log/syslog
May 29 11:16:02 ubuntu named[1234]: client 192.168.1.100#54321: query:
www.musicmania.in IN A +E (192.168.1.1)
May 29 11:16:02 ubuntu named[1234]: client 192.168.1.100#54321: query:
www.musicmania.in IN A +E (192.168.1.1)
student@ubuntu:~$ █
```

Look for errors such as:

- Zone file syntax error
- Permission denied
- Service failed to start

6) Verify Firewall (Optional)

```
student@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
53/tcp ALLOW Anywhere
53/udp ALLOW Anywhere
22/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
53 (v6) ALLOW Anywhere (v6)
53/udp (v6) ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)

student@ubuntu:~$ █
```

OR

```
student@ubuntu:~$ sudo ufw allow 53
Rules updated
Rules updated (v6)
student@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
53/tcp ALLOW Anywhere
53/udp ALLOW Anywhere
22/tcp ALLOW Anywhere

student@ubuntu:~$ █
```